

Overview:

The second exam focuses on modeling, supervised vs unsupervised learning, linear regression, model fit, residuals, and introductory text analysis (tokenizing, stopwords lemmatization, Zipf, stylometry, sentiment analysis). *This guide summarizes major ideas but does not list every detail covered in class.*

Supervised vs Unsupervised Learning

Difference between predicting an outcome (supervised) and finding patterns or groups (unsupervised).

Examples of each: regression, classification, clustering.

Identify whether a task is classification or regression.

Recognize what makes a problem supervised: the presence of a response variable y .

Keywords: supervised, unsupervised, prediction, clustering, classification, regression

Linear and Multilinear Regression

Understand the functional form $y = a + bx$ and $y = b_0 + b_1x_1 + b_2x_2 + \dots$

Interpret slopes and intercepts in plain language.

Explain what a linear relationship means and when it might fail (curved patterns).

Recognize how multiple predictors affect interpretation.

Understand what R^2 communicates about model fit.

Keywords: slope, intercept, predictor, response, linear relationship, R^2

Residuals and Model Fit

Define residuals as actual minus predicted values.

Interpret positive vs negative residuals.

Understand the goal of OLS: minimizing the sum of squared residuals.

Recognize good vs bad residual patterns (random cloud vs curvature).

Explain underprediction and overprediction using residuals.

Keywords: residual, prediction error, OLS, squared residuals, model fit

Training vs Testing

Understand why we split data: generalization to unseen cases.

Interpret differences between training error and testing error.

Explain why training data cannot be used for testing.

Keywords: train, test, generalization, overfitting, underfitting

Text Preprocessing

Tokenize text into words.

Remove punctuation and convert letters to lowercase.

Understand stopwords and why they are removed.

Identify what counts as a token after cleaning.

Keywords: token, tokenize, stopwords, lowercasing, punctuation removal

Sentiment Analysis

Use dictionaries of positive and negative words to evaluate a text's sentiment.

Merge tokenized words with sentiment dictionaries by matching on the word.

Count how many positive and negative words appear in each text or document.

Compute the proportion of positive and negative words out of all words.

Create bar charts to compare positive and negative sentiment across texts or authors.

Interpret what higher positive or negative proportions suggest about tone in context.

Keywords: sentiment, positive words, negative words, dictionary, proportion, bar chart

Zipf's Law and Word Frequencies

Explain frequency-rank relationship ($1, 1/2, 1/3, \dots$).

Compute simple proportional comparisons of word counts.

Interpret whether observed counts roughly follow Zipf.

Understand how Zipf appears across languages.

Keywords: Zipf's Law, rank, frequency, proportion, pattern

Stylometry and Author Fingerprints

Understand that authors have characteristic patterns in function-word usage.

Interpret frequency differences across authors.

Explain how an author's writing style can be used to classify a mystery text using word distributions.

Recognize how text statistics provide signatures.

Keywords: stylometry, function words, authorship, signature, frequency comparison

Basic Python for Modeling/Text

Select columns, filter rows, compute averages using pandas.

Apply operations to text columns (lowercase, split, ...).

Compute word frequencies using Python structures.

Understand DataFrames as rows (observations) and columns (variables).